

Improving mechanical performance for extreme environments

Prototype Project Report

WSU Mechanics and Materials lab



WSUM-PythonApps

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Project Requirements

I. Introduction

One of the many vital components to the space industry and defense is the protection against high velocity impact threats. It is crucial to know precisely how much damage manufactured protective shields can withstand before they break. Being able to accurately determine when a protective shield will fail will help these ultrahard ceramics and refractory alloys to be used safely and effectively to protect humans and equipment from impact threats.

Our team's aim is to develop a program that will analyze images of protective shields that have been damaged due to circumstances found in extreme environments, and then gather information about the shields. The main technique that we will be using is image segmentation. This will allow us to format the images in a way such that we will then be able to efficiently analyze them. By analyzing the damage as well as imperfections in the materials, we will be able to learn which circumstances allow for the best performance. This information will also be used in the development of stronger and more durable materials.

II. System Requirements Specification

II.1 Use Cases

1. Change Parameters

Pre-condition	- On main menu of User Interface
Post-condition	- Updated parameters of the image segmentation (pore identification) algorithm - Updated parameters of statistical analysis - Updated parameters of machine learning algorithm - Display new changes
Purpose	- Achieve accurate pore identification - Achieve accurate pore measurements - Achieve accurate correlation coefficient
Related Requirements	- Downloaded python executable

2. Import Images from Specified Folder

Pre-condition	- On main menu of the User Interface
Post-condition	- Imported image files into the current runtime - Display uploaded photos within folder - Display photos that were unable to be uploaded
Purpose	- Allows for images to be grouped for analysis

	- Gives analytic components of the app access to all necessary resources
Related Requirements	- Downloaded python executable - Images are locally downloaded in a folder on user's computer

3. Specify Groups of Imported Images for Analysis

Pre-condition	- On main menu of the User Interface - Images have already been imported
Post-condition	- Groups of images have been created - Display uploaded photos - Display photos that were selected but unable to be uploaded
Purpose	- Allows for images to be grouped for analysis - Gives analytic components of the app access to all necessary resources
Related Requirements	- Downloaded python executable - Images are locally downloaded on user's computer

4. Run Analysis on Groups

Pre-condition	- On main menu of the User Interface - Images have been successfully imported into Python program - Valid parameters were set for the group of images - Image groups have been created
Post-condition	- Display image segmentation results - Display statistical analysis results - Display parameters used for calculations - Display calculated results
Purpose	- Check if correct parameters were applied - Check if the image segmentation correctly identified pores - Identify patterns and relationships between pore structure and breakage patterns
Related Requirements	- Downloaded python executable - Images are locally downloaded on user's computer

5. Download Analytics to Computer

Pre-condition	- On main menu of the User Interface - Images have been successfully imported into Python program
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	- Valid parameters were set for group of images - Ran analysis on selected photos
Post-condition	- File of image segmentation, statistical and calculated results along with parameters is downloaded onto computer
Purpose	- Having a simple file to easily share with others and analyze results further - Document that shows information to replicate previous analysis on images
Related Requirements	- Downloaded python executable - Images are locally downloaded on user's computer

II.2 Functional Requirements

1. Graphical User Interface

Parameter Interface

Description	This interface will allow users to change parameters for the analysis of the images. This interface will allow parameters to be changed for the image segmentation, and statistical analysis.
Source	Internal requirements among members of the team
Priority	Priority level is medium. The ability to change parameters will increase the breadth of the applications use cases, but it is not a necessity. A further, and lower priority, development to this requirement would be to incorporate this ability into a GUI.

Select Images

Description	The ability to select images that the user would like to analyze. This will import and allow access for the python program to analyze
Source	External requirements requested by our client Dr. Zare
Priority	Priority level is high. The ability to select which images will be analyzed is essential to the functionality of this application. A fully integrated UI pop-up, however, is a low priority requirement that will make the application more user friendly but is not a necessity.

Download Results

Description	A button that will allow the user to download a file of the analysis.
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Source	External requirements requested by our client Dr. Zare
Priority	Priority level is medium for this functional requirement. It is a desired functionality that will provide exceptional use for the application; however, it is not an absolute necessity as some of the other requirements are.

2. Pore Identification

Description	Being able to identify and correctly label the pores of the material is crucial to calculating measurements and running statistical analysis on the data. Our approach is to use image segmentation to find these pores. Segmented images will be available to the user for further use if needed.
Source	External requirements requested by our client Dr. Zare
Priority	Priority level is high. This is essential and required functionality

3. Statistical Analysis

Description	Once we can accurately identify and measure the pores, we will run statistical analysis on the data to find any relationships between the pore structure and the failure point. Results will be visualized if possible.
Source	External requirements requested by our client Dr. Zare
Priority	Priority level is high. This is essential and required functionality

II.3 Non-functional Requirements

Ease-of-Use:

The system shall be simple enough for unfamiliar researchers to operate the program by selecting images for processing to obtain statistical feedback they want, without the need for any detailed instructions or tutorials.

Self-Contained:

The system shall be an enclosed program where there should be no need for the user to access or be redirected to outside resources or tools.

Efficient:

The system shall be an efficient alternative to manual calculations that relieves some of the burden off researchers to reduce the time and cost of the process, while compiling the data into a convenient and extensive resource.

III. System Evolution

Our current region-of-interest is the pores in the spall area, where our program will individually identify and measure each pore into a data set for analysis. While this is the focus of our assignment, we may also shift to different types of breakage patterns within the spall area depending on the client's research developments/needs potentially due to the data our program provides. This will require us to make sure our program is flexible in the image segmentation process in case a shift in focus to a different breakage pattern is desired, while keeping the statistical analysis feature consistent with such changes. Our team will keep in touch with Dr. Zare to stay updated on any developments in her research, as well as increasing our own understanding of the project domain.

Another fundamental aspect of our project to consider is the type of image segmentation. There are many different image segmentation techniques that we can use such as clustering and watershed and it may take extensive external research and testing on our end to determine the best method for the porous images. It is also important to consider the potential use of multiple image segmentation methods to obtain useful data.

II. Glossary

Image segmentation - The process of partitioning a digital image into multiple image segments.

III. References

[1]

“Image Segmentation,” *MathWorks*. <https://www.mathworks.com/discovery/image-segmentation.html#image-segmentation-with-matlab> (accessed Jan. 25, 2024).